Examples 11.38

Evaluate the following integrals

(i) $\int \frac{1}{(x-2)^2+1}dx$ (ii) $\int \frac{x^2}{x^2+5}dx$ (iii) $\int \frac{1}{\sqrt{1+4x^2}}dx$ (iv) $\int \frac{1}{\sqrt{4x^2-25}}dx$

Solution

(i) Let $I = \int \frac{1}{(x-2)^2+1}dx = \int \frac{1}{(x-2)^2+1^2}dx$

Putting $x - 2 = t \Rightarrow dx = dt$

Thus, $I = \int \frac{1}{t^2+1^2}dt = \tan^{-1}(t)+c = \tan^{-1}(x-2)+c$

(ii) Let $I = \int \frac{x^2}{x^2+5}dx$

$= \int \frac{x^2+5-5}{x^2+5}dx = \int\left(1-\frac{5}{x^2+5}\right)dx = \int dx - \int \frac{5}{x^2+5}dx$

$= x-5\int \frac{1}{x^2+\left(\sqrt{5}\right)^2}dx$

$= x-5\frac{1}{\sqrt{5}}\tan^{-1}\left(\frac{x}{\sqrt{5}}\right)+c$

$I = x-\sqrt{5}\tan^{-1}\left(\frac{x}{\sqrt{5}}\right)+c$

(iii) Let $I = \int \frac{1}{\sqrt{1+4x^2}}dx = \int \frac{1}{\sqrt{1+(2x)^2}}dx$

Putting $2x = t \Rightarrow 2\,dx = dt \Rightarrow dx = \frac{1}{2}dt$

Thus, $I = \frac{1}{2}\int \frac{1}{\sqrt{1^2+t^2}}dt$

$I = \frac{1}{2}\log\left|t+\sqrt{t^2+1}\right|+c = \frac{1}{2}\log\left|2x+\sqrt{(2x)^2+1}\right|+c$

$I = \frac{1}{2}\log\left|2x+\sqrt{4x^2+1}\right|+c$

(iv) Let $I = \int \frac{1}{\sqrt{4x^2-25}}dx = \int \frac{1}{\sqrt{(2x)^2-25}}dx$

Putting $2x = t \Rightarrow 2\,dx = dt \Rightarrow dx = \frac{1}{2}dt$

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